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Meta Review Status: The meta review and decision have not been released yet.



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Summary

This paper introduces an adaptive token-ordering approach for masked diffusion models (MDMs) and autoregressive models (ARMs), aiming to optimize inference efficiency by dynamically adjusting the token generation order. The core idea revolves around using reinforcement learning

(RL) to learn a policy that prioritizes the generation of easier tokens, thereby reducing computational cost and improving overall performance. The authors propose a novel framework where a π -learner, trained via entropy-regularized soft Q-learning, dynamically determines the token generation sequence based on the cumulative predictive V-information. This approach is motivated by the observation that not all tokens are equally difficult to predict, and by focusing on easier tokens first, the model can achieve better performance with less computation. The method is evaluated across a range of tasks, including structured reasoning puzzles, text generation, and downstream benchmarks like HumanEval and Math, demonstrating improvements in metrics such as perplexity, negative log-likelihood (NLL), and pass@1 accuracy. The authors introduce three adaptive inference oracles—vanilla, Top-K, and Margin—to further refine the token selection process. The empirical results suggest that the proposed adaptive token ordering strategy leads to significant reductions in perplexity and improvements in puzzle-solving accuracy compared to fixed or random token ordering approaches. The paper's main contribution lies in the novel application of reinforcement learning to the problem of token ordering, providing a dynamic and adaptive approach to sequence generation that is sensitive to the difficulty of individual tokens. The authors also provide a detailed formulation of the cumulative predictive V-information objective, linking token difficulty with inference order, which adds clarity to the proposed method. Overall, the paper presents a promising approach to optimizing inference in generative models, with potential implications for a wide range of applications. However, as I will discuss in detail, there are several areas where further investigation and clarification are needed to fully assess the method's practical applicability and robustness.

✓ Strengths

The primary strength of this paper lies in its innovative application of reinforcement learning to the problem of adaptive token ordering in generative models. The idea of dynamically adjusting the token generation sequence based on the difficulty of individual tokens, as measured by predictive V-information, is both novel and intuitively appealing. This approach offers a significant departure from traditional fixed-order generation methods, potentially leading to more efficient and effective inference. The authors' formulation of the cumulative predictive V-information objective provides a clear and principled way to quantify token difficulty, which is a key contribution of the work. Furthermore, the introduction of the π -learner, trained via entropy-regularized soft Q-learning, provides a concrete mechanism for implementing this adaptive token ordering strategy. The experimental results, while needing further scrutiny, demonstrate the potential of the proposed method. The reported improvements in perplexity, NLL, and task-specific metrics across various tasks, including structured reasoning puzzles and downstream benchmarks, suggest that the adaptive token ordering approach can indeed lead to better performance. The authors' evaluation across multiple tasks, including text generation and structured puzzles, with consistent improvements in perplexity, NLL, and task-specific metrics, is also a notable strength. This demonstrates the generalizability of the approach and its potential applicability to a wide range of problems. The use of three adaptive inference oracles—vanilla, Top-K, and Margin—further enhances the method's flexibility and effectiveness. Finally, the paper's focus on addressing the imbalance in subproblem difficulties during sequence generation is a significant contribution to the field, highlighting an important challenge in generative modeling and offering a creative solution.

✗ Weaknesses

Despite the promising aspects of this work, several significant weaknesses need to be addressed. First and foremost, the paper lacks a detailed analysis of the computational overhead introduced by the adaptive token ordering strategy. While the authors report FLOPs, they do not provide a breakdown of the time and memory costs associated with the π -learner and the dynamic token sequencing process. This is a critical omission, as the additional computational burden of the RL framework could potentially negate the benefits of adaptive token ordering, especially for very large models or datasets. The paper mentions the use of an entropy-regularized soft Q-learning loss (Equation 10) and alternating updates between the π -learner and the token predictor, which inherently add complexity and computational cost compared to standard training. However, there is no direct comparison of training time or resource consumption between the proposed method and standard training methods. This makes it difficult to assess the practical applicability of the method, especially in resource-constrained environments. My confidence in this weakness is high, as the paper's description of the joint training process and the use of reinforcement learning techniques clearly implies additional computational cost, and the absence of a direct comparison with standard training methods supports this concern. Second, the paper lacks detailed ablation studies on the impact of different hyperparameters, such as the entropy coefficient (α) and discount factor (γ). The authors provide specific values for these parameters ($\alpha = 0.1$, $\gamma = 1.0$) but do not explore how the model's performance changes with different hyperparameter settings. This is a significant weakness, as the performance of RL algorithms is often sensitive to hyperparameter choices.

Without a thorough ablation study, it is difficult to assess the robustness of the method and to determine the optimal configuration for different tasks and datasets. My confidence in this weakness is high, as the paper states the fixed values of α and γ without any exploration of their impact through ablation studies. Third, the paper's evaluation is limited by the choice of baselines. The authors primarily compare against fixed order ARMs and MDMs with random order, but do not include comparisons with other established token reordering or sampling techniques, such as beam search with different reordering strategies, or other non-autoregressive decoding techniques. This limits the ability to contextualize the performance gains achieved by the proposed approach. A more comprehensive comparison with a wider range of baselines would provide a better understanding of the method's advantages and limitations. My confidence in this weakness is high, as the limited set of baselines in Table 2 and the absence of comparisons with techniques like beam search with reordering are clearly evident. Fourth, the paper suffers from a lack of clarity in its presentation, particularly regarding the implementation of the reinforcement learning framework. The authors introduce Equation (10) as the core of their method, but fail to provide a clear explanation of how the state s_i and action a_i are defined within the context of token generation order. It is unclear how the state s_i is represented, whether it is the current sequence of tokens, the masked input, or some other representation. Similarly, the action a_i needs to be explicitly defined as the selection of the next token to be generated, including the specific mechanism for choosing among candidate tokens. Furthermore, the relationship between Equation (10) and the previously introduced predictive V-information, $I(y_i|X, y_{\setminus i})$



Suggestions

To address the identified weaknesses, I recommend several concrete improvements. First, the authors should conduct a thorough analysis of the computational overhead introduced by the adaptive token ordering strategy. This should include a detailed breakdown of the time and memory costs associated with the π -learner and the dynamic token sequencing process. A comparison of the inference time with and without the adaptive ordering, as well as an analysis of how these costs scale with sequence length and model size, is essential. Furthermore, the authors should provide a detailed analysis of the computational complexity of the π -learner, including the number of parameters and the FLOPs required for each inference step. This would allow for a more informed assessment of the practical applicability of the proposed method, especially in resource-constrained environments. It would also be beneficial to explore techniques to optimize the π -learner for faster inference, such as model quantization or pruning. Second, the authors should conduct a comprehensive ablation study on the impact of different hyperparameters, particularly the entropy coefficient (α) and discount factor (γ). This should include a systematic exploration of the hyperparameter space, using techniques such as grid search or more advanced hyperparameter optimization methods. The authors should analyze how these parameters affect the training dynamics and the final token ordering policy, including not only the final performance metrics but also the convergence speed and stability of the training process. This analysis should provide a more complete picture of the method's robustness and sensitivity to hyperparameter choices. Third, the authors should enhance the evaluation by including comparisons with a wider range of baseline models and token ordering strategies. This should include comparisons with established methods for token reordering or sampling, such as beam search with different reordering strategies, or other non-autoregressive decoding techniques. The evaluation should also include a variety of datasets and tasks to demonstrate the generalizability of the proposed method. Furthermore, the authors should provide a detailed analysis of the performance gains achieved by the proposed approach compared to these baselines, including statistical significance tests. Fourth, the authors should significantly improve the clarity of the paper's presentation, particularly regarding the implementation of the reinforcement learning framework. This should include a clear definition of the state s_i and action a_i in the context of token generation order, as well as a detailed explanation of how the reinforcement learning objective is connected to the minimization of predictive V-information. The authors should also provide a step-by-step explanation of how the π -learner is trained and how it interacts with the token predictor. Fifth, the authors should provide specific details about the model architectures used for the masked diffusion models (MDMs) and autoregressive models (ARMs). This should include the number of layers, the type of attention mechanisms, the dimensionality of the hidden states, and any other relevant architectural details. The authors should also clarify how the MDM and ARM interact with each other, including how the output of one model influences the input of the other. Sixth, the authors should provide a clear explanation of how the FLOPs are calculated, specifying which operations are included in the calculation and why these operations are relevant to the method's computational cost. Finally, the authors should ensure that all notations are clearly defined upon their first use. This includes providing explicit definitions for $I_{\setminus i}(\text{latent})$ and $I_{\setminus i}(\text{observation})$ in Table 1, explaining the meaning of P_i in the notation $P(y|X, y_{\setminus i})$.



Questions

Based on my analysis, I have several key questions that I believe would benefit from further clarification. First, how does the computational overhead of the adaptive token ordering strategy compare to fixed ordering approaches, particularly in terms of inference time and memory usage? I am particularly interested in a detailed breakdown of the costs associated with the π -learner and the dynamic token sequencing process, as well as how these costs scale with sequence length and model size. Second, could the authors provide more details on the sensitivity of the method to hyperparameter settings, particularly for α and γ , and how this might affect its performance across different tasks and datasets? I am interested in a more systematic exploration of the hyperparameter space and an analysis of how these parameters affect the training dynamics and the final token ordering policy. Third, are there any plans to extend this work to other types of generative models or tasks, and what challenges might arise in such extensions? I am curious about the generalizability of the proposed method and its potential applicability to a wider range of problems. Fourth, could the authors provide a more detailed explanation of the relationship between the reinforcement learning objective (Equation 10) and the predictive V-information? I am particularly interested in how the reward function is derived from the predictive V-information, or how the policy is trained to minimize it. Fifth, what specific architectures were used for the masked diffusion models (MDMs) and autoregressive models (ARMs), and how do these models interact with each other? I am interested in the details of the model architectures, including the number of layers, the type of attention mechanisms, and the dimensionality of the hidden states. Finally, how were the FLOPs calculated, and what specific operations are included in this calculation? I am interested in a more detailed explanation of the computational cost of the proposed method.

 Scores

Soundness: 2.75

Presentation: 2.25

Contribution: 2.5

Rating: 5.5

 AI Review From ZGCA will be automatically processed

 Summary

The paper proposes an adaptive token ordering framework for masked diffusion models (MDMs) and autoregressive models (ARMs). It formulates decoding order selection as a reinforcement learning (RL) problem that maximizes cumulative predictive V-information, $I_V(X \rightarrow$

$$y_i) = H_V(y_i | \emptyset) - H_V(y_i | X, y_)$$

✓ Strengths

- Clear, intuitive motivation: prioritize easier subproblems first using a token-difficulty metric (predictive V-information; Sec. 1–2).
- Novel formulation that links ordering decisions with cumulative predictive V-information optimized via RL (Sec. 4), plausibly contributing beyond fixed-order decoding.
- π -learner design with entropy-regularized soft Q-learning to encourage exploration among orderings (Sec. 4), theoretically appropriate for the sequential decision problem.
- Consistent scaling-law trends: validation NLL improves across compute and seeds (Table 4; Sec. 6), supporting the high-level premise.
- Empirical signals of downstream gains: perplexity reductions with minimal entropy change (Sec. 6), improved puzzle solve rates and pass@1 on HumanEval/Math (Sec. 6).
- Error imbalance analysis (Sec. 6) aligns with the motivation that not all tokens are equally difficult, supporting the need for adaptive ordering.

✗ Weaknesses

- Evaluation setup is under-specified and often non-standard: reliance on a 'simulated dataset' (Sec. 5), bespoke tests ('Sodomu MDM', 'Zebra MDM', 'L&O-NAE-SAT' in Sec. 6) with insufficient detail. Baselines such as the '42M ARM' are mentioned without architecture/training specifics or citations, hindering fair comparison and reproducibility.
- Key methodological details are missing: no pseudocode/algorithmic steps for the π -learner; the adaptive oracles (vanilla, Top-K, Margin) are central but not defined concretely (selection rules, thresholds, schedules, computational overhead). The Discussion introduces 'externally tuned oracle hyperparameters (such as the probability threshold η used in the sparse reward function)' without prior specification.
- Unclear how predictive V-information is estimated in practice: definitions of $H_V(\cdot)$ and its estimators are not provided; it is not explained whether NLL or calibration-adjusted surrogates are used, nor how estimation noise affects the reward signal (Sec. 4).
- Error imbalance computation relies on a 'stronger inference proxy' (Sec. 5–6) that is not described, limiting interpretability of the reported mean errors (0.7976 vs. 0.9724).
- Reported improvements (e.g., perplexity 60.0 → 52.0, HumanEval pass@1 60% → 66%) lack statistical rigor: no confidence intervals or significance tests; many results are single-point estimates (Sec. 6).
- Baselines and standard decoding strategies are not sufficiently compared: e.g., beam search, nucleus sampling, self-consistency, adaptive planning/blocked decoding. The very high perplexity baselines (≈ 60) raise concerns about the strength/representativeness of the underlying models and datasets.
- Compute, hardware, and resource usage are not documented; seed control is only mentioned for scaling experiments (Sec. 5). No code or dataset release is promised.
- No ablations on components (e.g., RL vs. heuristic ordering, effect of α/γ , reward shaping, oracle variants, cost of adaptivity) or qualitative error/failure analysis.

? Questions

- Predictive V-information: How exactly are $H_V(y_i|\emptyset)$ and $H_V(y_i|X, y_<i>$
- Reward specification: Is the RL reward the per-step estimated I_V , or a surrogate (e.g., negative NLL deltas)? How do you handle sparsity, normalization, and credit assignment over the sequence?
- π -learner details: Please provide pseudocode or detailed algorithmic steps for the soft Q-learning procedure (state/action definitions, target networks, replay buffers, on/off-policy setting, update schedules, temperature α annealing, stability tricks).
- Adaptive oracles: Precisely define 'vanilla', 'Top-K', and 'Margin' oracles in the context of token ordering. What are their selection rules, hyperparameters (including the ' η ' threshold mentioned in the Discussion), and compute overheads? How are K and margin tuned and scheduled at inference?
- Baselines and datasets: What datasets (public or proprietary) were used beyond the 'simulated dataset' (Sec. 5)? For puzzles ('Sodomu', 'Zebra', 'L&O-NAE-SAT'), please provide exact task definitions, dataset sizes/splits, generation protocols, and references. For the '42M ARM' baseline, specify architecture, training data, tokenization, and training steps.
- Standard decoding comparisons: How does your method compare to beam search, nucleus/top-p sampling, self-consistency/majority voting, remasking samplers, or blockwise decoding on the same models and datasets?
- Statistical rigor: Please provide multiple runs and confidence intervals for perplexity, entropy, solve rates, and pass@1 metrics. Are the improvements statistically significant?
- Error proxy: What is the 'stronger inference proxy' used to compute per-position errors (Sec. 5–6)? Please specify the method, its accuracy relative to ground truth, and why it is appropriate.
- Compute and reproducibility: Report hardware (GPUs, memory), training wall-time, FLOPs accounting details for the IsoFLOP sweep, and whether seeds were fixed across all experiments. Will you release code/checkpoints?
- Overhead and latency: What is the inference-time compute and latency overhead of adaptive ordering versus fixed ordering? Provide a breakdown as a function of sequence length and oracle type.
- Ablations: Could you isolate the contribution of (i) RL vs. heuristic (greedy difficulty), (ii) oracle choice, (iii) α and γ , (iv) reward shaping, and (v) teacher-forced sequence-specific orders (Sec. 6) to overall gains?
- Generalization: Do results hold on standard public corpora (e.g., WikiText, Penn Treebank, OpenWebText) and common reasoning benchmarks (e.g., GSM8K, Sudoku datasets with public splits)?

! Limitations

- The paper acknowledges sensitivity to oracle hyperparameters and fairness across distributions (Discussion), but does not quantify these effects or propose robust tuning procedures.
- Strong dependence on the base MDM/ARM quality: if base models are weak (as suggested by high perplexities), adaptive ordering gains may not generalize to competitive settings.
- Estimation noise in H_V/I_V could misguide the RL policy; no analysis of estimator calibration or robustness is provided.
- Inference-time overhead: adaptive ordering requires additional policy evaluation and oracle selection steps; the computational cost and latency trade-offs are not measured.
- Potential bias amplification: prioritizing 'easy' tokens may entrench model biases if 'ease' correlates with majority patterns; implications for fairness and minority patterns are not discussed.
- Non-standard tasks and proxies (e.g., 'Sodomu', 'Zebra', 'L&O-NAE-SAT', 'stronger inference proxy') limit external validity; missing comparisons on widely used public benchmarks.

- Limited discussion of negative societal impacts; adaptive decoding policies could be used to optimize for engagement or manipulation if deployed incautiously.

 Image Evaluation

Cross-Modal Consistency: 28/50

Textual Logical Soundness: 20/30

Visual Aesthetics & Clarity: 12/20

Overall Score: 60/100

Detailed Evaluation (≤ 500 words):

Visual ground truth (tables only):

- Table 1: Contains hyperparameters (α , γ , LR, batch size); title claims error statistics. Clear mismatch.
- Table 2: NLL vs FLOPs across seeds; decreasing trend with compute. Title claims ordering strategy comparison.
- Table 3: Hyperparameter configuration; matches content; readable.
- Table 4: NLL vs FLOPs across seeds; duplicates Table 2 values/trend.

1. Cross-Modal Consistency

- Major 1: Table 1 title does not match contents (error stats vs hyperparameters). Evidence: "Table 1: Empirical error statistics..." vs " α 0.1 Entropy regularization..."
- Major 2: Table 2 title claims ordering comparison but shows scaling with FLOPs. Evidence: "Table 2: Comparison of different ordering strategies..." and headers "Seed 1 $\times$ 10⁹ FLOPs..."
- Major 3: The text refers to error statistics summarized in Table 1, but none are present. Evidence: "Table 1 summarizes these error distributions..."
- Minor 1: Table 4 duplicates Table 2 with identical values without clarifying purpose. Evidence: "3.0248 ... -5.0117" appears in both tables.
- Minor 2: Exponent formatting inconsistent (" 1×10^9 FLOPs"). Evidence: " 1×10^9 FLOPs".
- Minor 3: Task names inconsistent ("Sudo/Sodomu"). Evidence: "Sudo and Zebra...", "The Sodomu MDM test..."

2. Text Logic

- Major 1: Key quantitative gains (perplexity 60 $\rightarrow$ 52, entropy 4.8 $\rightarrow$ 4.9; solve 70% $\rightarrow$ 80%) lack a dedicated figure/table, weakening verification. Evidence: "perplexity values drop from 60.0 ... to 52.0"
- Minor 1: Error metric definition mismatch ("absolute squared differences" vs $|e|$ in formula). Evidence: "absolute squared differences" vs " $\mu_{\text{lat}} = (1/N) \sum |e_i|$ "
- Minor 2: BCE-style SQL loss uses $\exp(Q)$ without stability discussion; unclear but not blocking. Evidence: " $-\exp(-Q) - (1 - \exp(-Q)) \cdot \log(1 - \exp(-Q))$ "

3. Figure Quality

- Major 1: Critical title–content mismatches (Tables 1–2) block intended messages. Evidence: See Cross-Modal Majors 1–2.

- Minor 1: Redundant tables (2 and 4) clutter narrative. Evidence: Identical NLL triplets across seeds.
- Minor 2: Minor typography (“109FLOPs”) reduces readability. Evidence: “1 × 109FLOPs”.

Key strengths:

- Clear, consistent NLL trends across seeds/compute suggest methodological effect.
- Method section presents coherent RL formulation and hyperparameters (Table 3).

Key weaknesses:

- Table title–content mismatches (1–2) critically undermine figure–text alignment.
- Central improvements (perplexity/solve rates) not tabulated or visualized.
- Minor notation/terminology inconsistencies (error metric, task names).

Recommendations:

- Fix Table 1 to actually report latent/observation error means/STD; move hyperparameters to Table 3 only.
- Rename Table 2 to “Validation NLL vs FLOPs” or supply the promised ordering comparison.
- Consolidate Tables 2 and 4 or differentiate (e.g., separate seeds or add error bars).
- Provide a results table for perplexity/entropy and puzzle/downstream metrics with clear oracle labels and seeds.
- Clarify error metric (absolute vs squared) and standardize task names.

 Scores

Originality: 3

Quality: 2

Clarity: 2

Significance: 2

Soundness: 2

Presentation: 2

Contribution: 2

Rating: 4 AI Review from SafeReviewer will be automatically processed **Summary**

This paper introduces a novel reinforcement learning (RL) framework designed to optimize token ordering during inference in masked diffusion models (MDMs) and autoregressive models (ARMs). The core idea revolves around dynamically adjusting the token generation sequence to prioritize easier subproblems, thereby improving overall inference efficiency and performance. The authors propose a π -learner, trained via entropy-regularized soft Q-learning, to determine the optimal token generation order based on the cumulative predictive V-information. This approach aims to reduce computational intractability associated with difficult token predictions. The paper presents three adaptive inference oracles—vanilla, Top-K, and Margin—to implement this dynamic ordering. The empirical evaluation demonstrates improvements across various tasks, including text generation, puzzle solving, and structured reasoning. Specifically, the authors report reductions in perplexity, enhancements in token diversity, and increased solve rates on structured puzzles. The paper also presents scaling law analyses, showing that the proposed method improves with increased computational resources. The authors argue that their approach bridges the gap between training-time hardness and inference-time adaptability, offering a promising direction for future research in generative modeling. However, the paper's presentation and experimental details require further refinement to fully support its claims and ensure reproducibility. The lack of detailed explanations of key components, such as the π -learner and the inference oracles, and the absence of a thorough analysis of computational overhead, limit the paper's impact. Additionally, the paper's reliance on a simulated dataset for the primary experiments raises concerns about the generalizability of the findings to real-world scenarios. Despite these limitations, the paper's core idea of using RL to optimize token ordering is innovative and has the potential to significantly improve the efficiency of generative models.

 Strengths

I find the core idea of using reinforcement learning to dynamically adjust token ordering during inference to be a significant strength of this paper. The approach of prioritizing easier subproblems, as defined by the cumulative predictive V-information, is a novel way to address the computational challenges associated with generating sequences using masked diffusion models and autoregressive models. The introduction of the π -learner, trained with an entropy-regularized soft Q-learning loss, is a technically innovative approach to learning the optimal token generation policy. The paper's empirical results, while needing further clarification, demonstrate the potential of this method to improve performance across a range of tasks. The reported reductions in perplexity, improvements in token diversity, and increased solve rates on structured puzzles suggest that the adaptive token ordering strategy can lead to tangible benefits. Furthermore, the inclusion of scaling law analyses, showing that the method improves with increased computational resources, adds to the credibility of the proposed approach. The authors' attempt to bridge the gap between training-time hardness and inference-time adaptability is a valuable contribution to the field of generative modeling. The exploration of different inference oracles (vanilla, Top-K, and Margin) also provides a useful perspective on how the adaptive ordering can be implemented in practice. Despite the weaknesses in presentation and experimental details, the core concept of using RL to optimize token ordering is a promising direction for future research and has the potential to significantly impact the efficiency of generative models.

Weaknesses

After a thorough examination of the paper, I have identified several significant weaknesses that impact its overall quality and credibility. First and foremost, the paper suffers from a lack of clarity and precision in its writing. There are numerous instances of undefined variables and insufficient explanations of key concepts. For example, in Section 2, the term 'Hv' in the equation for predictive V-information is not defined, making it difficult to understand the specific measure of uncertainty being used. Similarly, the term 'V' in the cumulative predictive V-information is not explicitly defined within the equation context, even though it is defined earlier in the text. This lack of precision hinders the reader's ability to fully grasp the proposed method. Furthermore, the paper uses the term 'entropy' without a clear definition of what it refers to, specifically whether it is per-position entropy or sequence entropy. This ambiguity makes it difficult to interpret the reported entropy values and their implications. The paper also lacks a clear explanation of the π -learner's architecture and training process. While the paper introduces the π -learner as a novel component, it does not provide sufficient details on its internal workings, making it difficult to understand how it learns to determine the optimal token generation order. The paper also fails to provide a clear explanation of the three adaptive inference oracles (vanilla, Top-K, and Margin). The descriptions of these oracles are vague, and the paper does not provide sufficient details on how they operate and how they differ from each other. This lack of clarity makes it difficult to assess the practical implications of the proposed method. The experimental section also suffers from several weaknesses. The paper relies on a 'simulated dataset' for the primary experiments, but it does not provide sufficient details on how this dataset was generated. This lack of information makes it difficult to assess the generalizability of the findings to real-world scenarios. The paper also lacks a thorough analysis of the computational overhead associated with the proposed method. While the paper reports performance improvements, it does not provide any information on the inference time or computational cost of the adaptive token ordering strategy. This omission makes it difficult to assess the practical feasibility of the method. The paper also presents results on various benchmarks, but it does not provide sufficient details on the specific tasks used within each benchmark. For example, the paper mentions 'HumanEval, Math, MMLU, and ROCStories' but does not specify the exact tasks within these benchmarks. This lack of detail makes it difficult to interpret the reported results. Additionally, the paper's presentation of results is inconsistent and difficult to follow. The paper presents results for different oracles without always explicitly stating which oracle was used for each specific result. This lack of clarity makes it difficult to compare the performance of different oracles and to assess the overall effectiveness of the proposed method. The paper also lacks a discussion of the potential limitations of the proposed approach. For example, the paper does not discuss the potential for error propagation when using adaptive token ordering. This omission makes it difficult to assess the robustness of the method. Finally, the paper's reliance on a single, small table (Table 1) for presenting empirical error statistics is insufficient to fully support the claims about error imbalances. The lack of detailed analysis and visualization of these errors limits the paper's impact. The paper also lacks a clear explanation of the connection between the RL framework and the observed error imbalances. The paper states that the RL framework is motivated by the error imbalances, but it does not clearly explain how the RL framework addresses these imbalances. In summary, the paper's weaknesses stem from a lack of clarity, insufficient experimental details, and a lack of thorough analysis, which significantly impact the paper's overall quality and credibility. The lack of precise definitions, the use of a simulated dataset without sufficient details, the absence of computational overhead analysis, and the inconsistent presentation of results all contribute to these weaknesses. The lack of a clear explanation of the π -learner and the inference oracles further limits the paper's impact. These issues need to be addressed to improve the paper's clarity, reproducibility, and overall contribution to the field.

Suggestions

To address the identified weaknesses, I recommend several concrete and actionable improvements. First, the authors should significantly improve the clarity and precision of their writing. This includes providing clear definitions for all variables and terms used in the paper, especially within equations. For example, the authors should explicitly define 'Hv' and clarify what type of entropy 'H' represents in Section 2. They should also provide a clear definition of 'V' within the equation context. The authors should also clarify whether the entropy referred to is per-position or sequence entropy. Furthermore, the authors should provide a more detailed explanation of the π -learner's architecture and training process. This should include a description of the input and output spaces, the specific neural network architecture used, and the training algorithm. The authors should also provide a clear explanation of the three adaptive inference oracles (vanilla, Top-K, and

Margin), including how they operate and how they differ from each other. This should include a discussion of the specific algorithms used for each oracle and the parameters involved. Second, the authors should provide more details on the experimental setup. This includes providing a detailed description of how the 'simulated dataset' was generated, including the size of the vocabulary, the length of the sequences, and the specific simulation process. The authors should also provide a thorough analysis of the computational overhead associated with the proposed method. This should include a comparison of the inference time and computational cost of the adaptive token ordering strategy with a standard, non-adaptive approach. The authors should also provide more details on the specific tasks used within each benchmark. For example, they should specify the exact tasks within HumanEval, Math, MMLU, and ROCStories. This will allow for a better understanding of the reported results. The authors should also present their results in a more consistent and organized manner. This includes clearly labeling each result with the corresponding oracle used and providing a summary table that compares the performance of different oracles across all tasks. The authors should also include a discussion of the potential limitations of the proposed approach, such as the potential for error propagation when using adaptive token ordering. This should include an analysis of how the adaptive ordering might affect error propagation compared to fixed ordering strategies. The authors should also provide a more detailed analysis of the error imbalances, including visualizations of the errors across different positions. This should include a discussion of the specific types of errors that are more likely to occur at different positions. The authors should also clarify the connection between the RL framework and the observed error imbalances. This should include an explanation of how the RL framework is designed to address these imbalances. Finally, the authors should consider including a more detailed discussion of the potential benefits of their approach for non-autoregressive generation. This should include a discussion of how the adaptive token ordering could be used to improve the performance of non-autoregressive models. By addressing these weaknesses, the authors can significantly improve the clarity, reproducibility, and overall contribution of their paper.

? Questions

After reviewing the paper, I have several questions that I believe are crucial for a deeper understanding of the proposed method and its implications. First, I am curious about the specific architecture of the π -learner. While the paper introduces it as a novel component, it lacks a detailed explanation of its internal workings. Could the authors provide a more detailed description of the π -learner's architecture, including the specific neural network layers used, the input and output representations, and the training process? Second, I am interested in the practical implications of the adaptive token ordering strategy. The paper mentions three adaptive inference oracles (vanilla, Top-K, and Margin), but it does not provide sufficient details on how they operate and how they differ from each other. Could the authors provide a more detailed explanation of these oracles, including the specific algorithms used for each oracle and the parameters involved? Third, I am concerned about the generalizability of the findings. The paper relies on a 'simulated dataset' for the primary experiments, but it does not provide sufficient details on how this dataset was generated. Could the authors provide more information on the simulation process, including the size of the vocabulary, the length of the sequences, and the specific parameters used? Fourth, I am curious about the computational overhead associated with the proposed method. While the paper reports performance improvements, it does not provide any information on the inference time or computational cost of the adaptive token ordering strategy. Could the authors provide a more detailed analysis of the computational overhead, including a comparison of the inference time and computational cost of the adaptive token ordering strategy with a standard, non-adaptive approach? Fifth, I am interested in the specific tasks used within each benchmark. The paper mentions 'HumanEval, Math, MMLU, and ROCStories' but does not specify the exact tasks within these benchmarks. Could the authors provide more details on the specific tasks used within each benchmark? Finally, I am curious about the connection between the RL framework and the observed error imbalances. The paper states that the RL framework is motivated by the error imbalances, but it does not clearly explain how the RL framework addresses these imbalances. Could the authors provide a more detailed explanation of how the RL framework is designed to address these imbalances? These questions are crucial for a deeper understanding of the proposed method and its implications, and I believe that addressing them would significantly improve the paper's overall quality and credibility.

Scores

Soundness: 2.0

Presentation: 1.75

Contribution: 2.25

Rating: 3.75

Keywords

Extract Keywords

Click the button to extract keywords

Insights

Insights Extracted

- 

Formulating token generation ordering as a reinforcement learning problem optimized by cumulative predictive V-information enables dynamic adaptation that prioritizes easier tokens first, significantly reducing error propagation and improving sequence generation performance.
- 

The introduction of a novel n-learner combined with multiple adaptive inference oracles (vanilla, Top-K, and Margin) substantiates that adaptive token ordering consistently lowers perplexity and enhances task-specific metrics, including structured puzzle-solving and downstream benchmarks like HumanEval and Math.
- 

Adaptive token ordering bridges the gap between training-time token difficulty and inference-time flexibility, establishing a scalable and robust framework that preserves output diversity while addressing heterogeneous token prediction challenges across masked diffusion and autoregressive generative models.